

# CBMS

Central Biorepository Management Software

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## User Manual

Version 1.0.0 · 2026

*For internal use by authorised research staff only.*

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# 1. Overview

CBMS (Central Biorepository Management Software) is a local desktop application for managing the full lifecycle of biological samples in HIV and infectious-disease research cohort studies. It runs entirely on a single workstation with no internet connection required.

## What CBMS manages

- **Participants** — demographic registration and visit records
- **Samples & Aliquots** — collection, typing, and aliquot tracking
- **Storage** — 6-level freezer hierarchy down to individual box positions
- **Shipments** — shipping aliquots to researchers with full history
- **Catalogue** — pivot-table view of available sample inventory
- **Audit trail** — every action logged with timestamp and user
- **User access control** — three roles with distinct permissions

## Tab layout

| Tab          | Purpose                                 | Keyboard shortcut |
|--------------|-----------------------------------------|-------------------|
| Dashboard    | KPI summary + cohort flowchart          | Ctrl+1            |
| Studies      | Create and manage study projects        | Ctrl+2            |
| Participants | Register and search participants        | Ctrl+3            |
| Samples      | Add samples and aliquots per visit      | Ctrl+4            |
| Storage      | Visual box grid — place & move aliquots | Ctrl+5            |
| Search       | Multi-criteria search across all data   | Ctrl+F            |
| Shipments    | Create and track shipments              | Ctrl+6            |
| Catalogue    | Sample-type pivot table & export        | Ctrl+7            |
| Admin        | Users, audit log, backup                | Ctrl+8            |

## 2. Installation

### macOS

Requirements: macOS 11 or later, Python 3.10+ (only needed to build; the distributed app bundle is self-contained).

#### Option A — Run the installer script (recommended)

- Open Terminal.
- Navigate to the CBMS project folder:
- Run the installer:

```
cd /path/to/cbms_with_excel_import_clean
chmod +x install_and_build.sh
./install_and_build.sh
```

The script will install all dependencies, build **dist/CBMS.app**, and create **dist/CBMS-v1.0.0.dmg**.

#### Option B — Install from DMG

- Double-click **CBMS-v1.0.0.dmg**.
- Drag **CBMS** to the **Applications** folder.
- Eject the disk image.
- Right-click CBMS in Applications → **Open** → click **Open** when macOS warns about an unsigned app.

■ After the first successful open, the app launches normally by double-clicking.

### Windows

Requirements: Windows 10 or later, Python 3.10+ (only needed to build).

#### Option A — Run the installer script

- Double-click **install\_and\_build.bat**.
- The script installs all dependencies and produces **dist\CBMS.exe**.
- Copy **CBMS.exe** to any location and double-click to run.

#### Option B — Run without building

```
venv\Scripts\activate
python main.py
```

### First Launch

On first launch CBMS automatically creates the database at **data/cbms.db** and seeds a default administrator account:

| Field    | Default value |
|----------|---------------|
| Username | admin         |
| Password | Admin@1234    |

■■ Change the admin password immediately after first login via Admin → Users.

## 3. Login & User Roles

Enter your username and password at the login screen. Contact your system administrator if you need an account created.

### Roles and permissions

| Role     | Who holds it           | Key permissions                                                          |
|----------|------------------------|--------------------------------------------------------------------------|
| PI       | Principal Investigator | Full access — create studies, manage users, delete records, ship samples |
| Manager  | Study Coordinator      | Register participants, import data, block/ship aliquots, manage users    |
| Lab Tech | Bench Technician       | Add samples, place aliquots in storage, unblock aliquots, run searches   |

■ ■ Buttons requiring a higher permission are disabled (greyed out), not hidden. This lets all users see what operations exist.

## 4. Dashboard

The Dashboard opens on login and gives a real-time snapshot of the biorepository.

### KPI strip

Six numbers are displayed across the top:

- **Participants** — total registered across all studies
- **Samples** — total sample records
- **Aliquots** — total aliquot tubes
- **Available** — aliquots currently available for use
- **Blocked** — aliquots reserved for a researcher
- **Shipped** — aliquots included in completed shipments

### Cohort flowchart

Below the KPI strip is a cohort workflow table. Use the **Cohort** dropdown (top-right of the flowchart area) to select a specific cohort. The table shows participant counts and vial counts per visit and sample type.

## 5. Studies Tab

A Study (or Project) is the top-level container. All participants, samples, and visit definitions belong to a study.

### Creating a study

- Click **New Study**.
- Enter the **Project ID** (short code, e.g. COH) — this prefixes all sample IDs.
- Enter the **Project Name** and optional description.
- Click **Save**.

■ The Project ID cannot be changed after creation. Choose it carefully.

### Visit definitions

Each study has visit definitions (Screening, Enrollment, Follow-up). These must exist before importing samples via Excel.



## 6. Participants Tab

The Participants tab lists all registered participants with search and filter controls.

### Registering a participant manually

- Select a study from the **Study** dropdown.
- Click **Register Participant**.
- Fill in PID, Age, Gender, Population, Disease, Site, Cohort Name.
- Click **Save**.

### Importing participants from Excel

Use the Excel bulk import to load hundreds of rows at once.

- Click **Import from Excel**.
- Select the study to import into.
- Choose your **.xlsx** file (must follow the 21-column format — see Section 13).
- Review the validation summary. Fix any flagged rows before proceeding.
- Click **Import** to commit all rows in one transaction.

■ ■ If any row fails validation the entire import is cancelled. No partial imports are written.

### Searching and filtering

- Use the **Study** dropdown to filter by study.
- Type in the **PID** search box for partial matches.
- Results are paginated (100 per page) — use **Prev / Next** to navigate.

### Exporting participants

Click **Export to Excel** to save all filtered participants (all pages, not just the visible page) to an **.xlsx** file.

## 7. Samples Tab

The Samples tab shows samples and aliquots for selected participant(s).

### Selecting participants

- Choose a **Study** from the top dropdown.
- The participant list loads sorted alphabetically (A→Z).
- Type in **Search PID...** to filter the list in real time.
- Check one or more participant checkboxes — their samples merge into the right panel.
- Use **Select All** / **Clear All** for quick batch selection.

### Visit filter

Click a visit code in the left **Visits** list to filter samples to that visit. Click **All Visits** to reset.

### Adding a sample

- Select exactly **one** participant checkbox.
- Click **■ Add Sample**.
- Choose Sample Type, Visit, Date Collected, and optional discrepancy notes.
- Click **Save**.

### Adding aliquots to a sample

- Select a sample row in the right panel.
- Click **■ Add Aliquots**.
- Specify quantity and volume (µL) per aliquot.
- Click **Save**.

### Aliquot status colours

| Colour | Status    | Meaning                          |
|--------|-----------|----------------------------------|
| Green  | Available | Ready for use or shipment        |
| Orange | Blocked   | Reserved for a researcher        |
| Blue   | Shipped   | Included in a completed shipment |

## 8. Storage Tab

The Storage tab provides a visual map of every freezer, shelf, rack, drawer, and box. Click a box node in the left tree to see its 10×10 position grid on the right.

### Storage hierarchy

| Level          | Examples                          | Notes                       |
|----------------|-----------------------------------|-----------------------------|
| Freezer / Tank | NARI/COHRPICA/18-19/01<br>REGULAR | Upright or cylindrical      |
| Compartment    | I, II, III, IV (Shelf)            | Upright freezers only       |
| Rack           | A, B, C, D, E, F                  | Upright freezers only       |
| Drawer         | 01, 02, 03, 04, 05                | Upright freezers only       |
| Box            | Container name from Excel col O   | Holds 10×10 = 100 positions |
| Position       | A1 – J10                          | Individual tube slot        |

### Box grid colours

| Colour | Meaning                                 |
|--------|-----------------------------------------|
| White  | Empty — no aliquot placed               |
| Blue   | Occupied — aliquot present              |
| Orange | Blocked — aliquot reserved              |
| Grey   | Shipped — aliquot sent; position locked |
| Green  | Selected — currently clicked cell       |

### Placing an aliquot

- Select a box in the hierarchy tree.
- Click an empty (white) cell in the grid.
- Click **Place Aliquot**.
- Search for the aliquot by ID and confirm.

### Moving an aliquot

- Click the occupied cell.
- Click **Move** or drag the cell to a new empty position.

■ Shipped (grey) cells are locked — Move and Remove are disabled.

## 9. Search Tab

The Search tab lets you find aliquots using any combination of participant, sample, and storage criteria. Use **Ctrl+F** to open it instantly.

### Filter fields (left panel)

| Filter      | Match type                        |
|-------------|-----------------------------------|
| PID         | Partial text match                |
| Population  | Partial text match                |
| Age         | Exact integer                     |
| Site        | Partial text match                |
| Visit Time  | Partial text match (e.g. M0, SCR) |
| Visit Code  | Partial text match (e.g. 1.0)     |
| Cohort      | Partial text match                |
| Disease     | Partial text match                |
| Sample Type | Partial text match                |

Use the **AND / OR** toggle to combine filters. Results are paginated at 100 per page.

### Actions on results

| Button              | Description                                                | Who can use  |
|---------------------|------------------------------------------------------------|--------------|
| Show in box         | Switch to Storage tab and highlight the aliquot's position | All          |
| Block selected...   | Reserve aliquots for a researcher until a date             | Manager / PI |
| Unblock selected... | Release a block with a mandatory reason                    | All roles    |
| Ship selected...    | Add aliquots to a new or existing shipment                 | Manager / PI |
| Export to Excel     | Download current search results as .xlsx                   | All          |

■ Double-click any row to jump directly to the aliquot's position in the Storage tab.

## 10. Shipments Tab

The Shipments tab records aliquots sent to external researchers. Once shipped, an aliquot's box cell turns grey and the aliquot cannot be moved or removed.

### Creating a shipment

- Go to the **Search** tab and search for the aliquots to ship.
- Select the rows (hold Ctrl or Shift for multiple).
- Click **Ship selected...**
- Fill in researcher name, institution, and shipment date.
- Click **Ship** — a unique shipment reference (SHIP-YYYYMMDD-NNN) is generated.

### Viewing shipment history

Open the **Shipments** tab to see all shipments. Click a shipment row to see the list of aliquots it contained.

## 11. Catalogue Tab

The Catalogue tab generates a pivot table showing how many aliquots of each sample type each participant has.

### Generating the catalogue

- Select a **Study** (or leave at 'All studies').
- Optionally check **Available aliquots only** to exclude shipped/blocked.
- Click **Generate Catalogue**.

### Narrowing results

Use the **Narrow results** filters (PID, Gender, Disease, Cohort, Site, Sample Type) to slice the pivot table without regenerating it.

### Exporting

Click **Export to Excel...** to save the current pivot view to an .xlsx file.

## 12. Admin Tab

The Admin tab is restricted to PI and Manager roles. It has three sub-tabs.

### Users

- **Add user** — click  Add User, enter username, role, email, and temporary password.
- **Edit user** — select a row and click  Edit to change name, email, or role.
- **Reset password** — select a user and click  Reset Password.
- **Deactivate** — select a user and click  Deactivate (does not delete records).

  A user cannot change their own role. Role changes must be made by another PI or Manager.

### Audit Trail

Every significant action is logged. The audit trail is immutable — entries cannot be edited or deleted.

- Filter by **Action** type (CREATE, UPDATE, DELETE, LOGIN, SHIP, BLOCK, UNBLOCK...).
- Filter by **Entity type** and **date range**.
- Click **Search** to refresh results (100 per page).
- Click column headers to sort.

### Backup

The current backup timestamp is shown at the top of the Admin tab and in the status bar at the bottom of the window.

- Click  **Backup Now** (in the Admin tab header) or press **Ctrl+B** from anywhere.
- A timestamped copy is saved to **data/backups/cbms\_YYYYMMDD\_HHMMSS.db**.

 Take a backup before every Excel import and before any major data change.

## 13. Excel Import Reference

The Excel file must contain exactly 21 columns in this order. The header row is validated (case-insensitive, minor typos accepted).

| C<br>ol | Header             | Required | Notes                                                                                      |
|---------|--------------------|----------|--------------------------------------------------------------------------------------------|
| A       | PID                | Yes      | Participant ID string                                                                      |
| B       | Age                | No       | Integer                                                                                    |
| C       | Gender             | No       | Male / Female / Transgender                                                                |
| D       | Population         | No       | FSW, MSM, PWID, General Adult, Child only, Pair-Child, Pair-Mother                         |
| E       | Disease            | No       | None, Diabetes, TB, Risk of CVD, Infected without co-morbidity, Unknown-Screen failure, NA |
| F       | Visit Code         | No       | Decimal (e.g. 1.0)                                                                         |
| G       | Visit Time         | No       | SCR, M0 – M36                                                                              |
| H       | Date Collected     | No       | Any parseable date format                                                                  |
| I       | Site Name          | No       | GHTM, ICMR-NARI (or NARI), NIMHANS, NIRT, YRG-Care                                         |
| J       | Visit Name         | No       | Screening, Enrollment, Follow-up                                                           |
| K       | Sample Type        | No       | Serum, ED Plasma, HEP Plasma, EDTA PBMC                                                    |
| L       | Cohort Name        | No       | HIV UNINFECTED, HIV INFECTED-ADULT, HIV INFECTED-PEDIATRIC, EARLY HIV INFECTED             |
| M       | Aliquot ID         | No       | Auto-generated if blank                                                                    |
| N       | Freezer / Tank     | Cond     | Required if any storage column is filled                                                   |
| O       | Container          | Cond     | Box name                                                                                   |
| P       | Slot Position      | Cond     | 1–100 sequential position in box                                                           |
| Q       | Shelf              | Cond     | I / II / III / IV (upright); '.' for cylindrical                                           |
| R       | Rack               | Cond     | A-01 format (upright); 1–13 (cylindrical)                                                  |
| S       | Position           | N/A      | Auto-computed from Slot Position; use '.'                                                  |
| T       | Discrepancy Remark | No       | Free text or '.'                                                                           |
| U       | Discrepancy For    | No       | Free text or '.'                                                                           |

■ If any storage column (N–S) is non-empty, all of N, O, P, Q, R must be filled. Partial storage paths are a validation error.



## 14. Backup & Recovery

### Taking a backup

- Press **Ctrl+B** from anywhere in the app, or
- Open **Admin** tab and click **Backup Now**.
- The backup is saved to **data/backups/**.
- The status bar (bottom-right) and Admin tab header show the last backup time.

### Restoring from backup

- Close CBMS.
- Navigate to **data/backups/** and find the desired **.db** file.
- Copy it to **data/cbms.db** (overwriting the current database).
- Relaunch CBMS.

■■ Restoring a backup permanently replaces the current database. Take a backup of the current state first if in doubt.

### Backup file location

```
data/  
  cbms.db           ← live database  
  backups/  
    cbms_20260514_143022.db  
    cbms_20260513_090011.db  
    ...
```

## 15. Keyboard Shortcuts

| Shortcut     | Action                         |
|--------------|--------------------------------|
| Ctrl+1       | Dashboard                      |
| Ctrl+2       | Studies                        |
| Ctrl+3       | Participants                   |
| Ctrl+4       | Samples                        |
| Ctrl+5       | Storage                        |
| Ctrl+F       | Search                         |
| Ctrl+6       | Shipments                      |
| Ctrl+7       | Catalogue                      |
| Ctrl+8       | Admin                          |
| Ctrl+N       | New participant (from any tab) |
| Ctrl+Shift+N | New study                      |
| Ctrl+B       | Backup now                     |

## 16. Troubleshooting

| Problem                                                  | Solution                                                                                                                                                                        |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| App won't open on macOS (unsigned warning)               | Right-click CBMS.app → Open → click Open. Or run:<br><code>xattr -d com.apple.quarantine /Applications/CBMS.app</code>                                                          |
| Excel import fails — header mismatch                     | Check that your file has exactly 21 columns in the correct order. Minor spelling variants (e.g. 'Descripancy') are accepted.                                                    |
| Excel import fails — validation errors                   | The error list shows the row number and field. Fix those rows in Excel and retry.                                                                                               |
| Search by cohort returns no results                      | Ensure the Cohort filter text matches the stored value (e.g. 'HIV UNINFECTED'). The filter is a partial case-insensitive match.                                                 |
| Box grid shows no aliquots after import                  | Check that the Slot Position column (P) contains a number 1–100 and that all storage columns (N–R) are filled.                                                                  |
| 'No storage location' when double-clicking Search result | The aliquot was imported without storage coordinates or was removed from its position.                                                                                          |
| Forgot admin password                                    | Close the app, delete data/cbms.db, and relaunch — the default admin/Admin@1234 will be restored. Warning: this also deletes all data. Restore from backup instead if possible. |